

Syllabus

Department	Math & CS	Credits	3	Instructor	Sanghyun Yoon	Class Room	5708
Subject	MC4206: Algorithms	Class hrs/wk	3	Lab (e-mail)		Attendee	
				5707			

1 Course Description

Course Objectives

On completion of this course, you'll hopefully be able to:

- Model/analyze real-world problems and design algorithmic solutions along with mathematical proofs using discrete structures and combinatorial methods.
- Understand/appreciate limits of computation.

Contents

- Intractability (algorithm design anti-patterns)
 - NP-completeness, Karp reduction vs. Cook/Turing reduction, co-NP
 - PSPACE-completeness, #P-completeness
- Algorithm design/analysis patterns for tractable problems
 - Greedy strategy (matroids), divide-and-conquer (e.g. FFT)
 - Dynamic programming (convex hull trick, D&C opt., Knuth opt.)
 - Polynomial-time reduction (e.g. max-flow/min-cut, 2-SAT)
 - Randomization (ZPP, RP, BPP, derandomization methods)
 - Amortized analysis (e.g. disjoint-set data structure)
- Algorithm design patterns for intractable problems
 - Subproblem analysis (e.g. fixed-parameter tractability)
 - Approximation (e.g. pricing, LP rounding, rounding/trimming for FPTAS)
- Algorithms for online problems
- Quantum algorithms and quantum-resistant cryptography
 - Quantum computation, BQP
 - Shor's algorithms for discrete logarithm and factoring
 - Lattice-based public-key encryption schemes (learning with errors)
- Generic algorithms for mathematical programming
 - Simplex method for LP, LP duality, ellipsoid method (separation oracle)
 - Interior point method, gradient descent for convex optimization
 - Cutting-plane method, branch-and-cut method for integer linear programming

2 Text & References

Course online: <https://classroom.google.com/u/1/c/ODQzNjc0MTExMzAw>

- “Algorithm Design”, J. Kleinberg and E. Tardos
- “Computers and Intractability: A Guide to the Theory of NP-Completeness”, M. Garey et al.
- “Introduction to Theory of Computation”, M. Sipser
- “Introduction to Algorithms”, T. Cormen, C. Leiserson, R. Rivest, and C. Stein
- “Algorithms”, S. Dasgupta, C. H. Papadimitriou, and U. V. Vazirani
- “Introduction to Linear Optimization”, D. Bertsimas and J. Tsitsiklis
- “Linear Programming: Foundations and Extensions”, R. J. Vanderbei
- “Discrete Optimization”, G. Parker and R. Rardin
- “Convex Optimization”, S. Boyd and L. Vandenberghe
- “Approximation Algorithms”, U. V. Vazirani
- “Online Computation and Competitive Analysis”, A. Borodin and R. El-Yaniv
- “Computational Complexity”, S. Arora and B. Barak
- “Quantum Computation and Quantum Information”, M. A. Nielsen and I. L. Chuang
- “An Intensive Introduction to Cryptography”, B. Barak
- “A Decade of Lattice Cryptography”, C. Peikert
- “Combinatorial Optimization: Polyhedra and Efficiency”, A. Schrijver
- “Discrete and Combinatorial Mathematics”, R. Grimaldi

3 Grading

- Grading table

Activities	Percentages
Problem Sets	10%
Final exam	80%
Attendance	10%

- 15 problem sets
 - Scoring policy: full score if 2/3 submitted, otherwise 0 point
- Final exam: 9:00-23:00 on the Saturday right before the final exam period (**2026/6/13**)
- Make-up final exam for officially approved absence (e.g. COVID, illness, out-of-school competition, conference)
 - immediately after the day all students are released from quarantine/...
 - 0 point if do not show up on the make-up final test
- Late-work policy: not accepted

4 Lecture Schedule

- 1-7: Intractability (7)
- 8: 2-SAT (1)
- 9-12: Flow Network (4)
- 13-15: Greedy strategy (3)
- 16-17: Divide-and-conquer (2)
- 18-20: Dynamic programming (3)
- 21: Amortized analysis (1)
- 22-27: Linear/mathematical programming (6)
- 28-29: Subproblem analysis (2)
- 30-32: Approximation algorithms (3)
- 33-34: Randomized algorithms (2)
- 35-36: Online algorithms (2)
- 37-40: Quantum algorithms & quantum-resistant cryptography (4)

Intractability (7)

1. Polynomial-time reductions #1
 - Algorithmic problems: decision vs. search vs. optimization
 - Polynomial-time reductions
 - Standard reduction techniques (simple equivalence, restriction, local replacement, component design)
2. Polynomial-time reductions #2
 - 3-SAT vs. 2-SAT
 - Karp reduction vs. Cook/Turing reduction
 - Self-reducibility
 - Anatomy of poly-time reductions: a certifier-based perspective
3. Reduction examples #1
 - $3\text{-SAT} \leq_p 3\text{-dimensional matching}$
 - $3\text{-SAT} \leq_p \text{directed Hamiltonian cycle} \leq_p \text{Hamiltonian cycle} \leq_p \text{TSP}$
 - $3\text{-SAT} \leq_p \text{longest simple path}$
4. Reduction examples #2
 - $3\text{-SAT} \leq_p 3\text{-colorability} \leq_p 3\text{-colorability on planar graphs}$
 - Planar k -colorability
 - $3\text{-SAT} \leq_p \text{subset sum}$
 - $3\text{-DM} \leq_p \text{subset sum}$
 - $\text{Subset sum} \leq_p \text{partition}$
 - $\text{Subset sum} \leq_p \text{scheduling with release time}$
5. NP-completeness

- Certifier-based definition of the class NP
- Cook-Levin theorem
- NP-easy vs. NP-hard vs. NP-complete
- How to cope with intractability
- Complexity classes beyond NP (co-NP, PSPACE, ZPP, BPP, RP, #P, PPAD, BQP)

6. co-NP

- Asymmetry of NP
- Problems in $NP \cap co-NP$

7. PSPACE

- Quantified satisfiability
- Planning problems
- PSPACE-completeness

2-SAT (1)

8. 2-SAT

- Kosaraju's algorithm for identifying strongly connected components (SCCs)
- Implication graph for 2-CNF, algorithm for 2-SAT
- Reductions to 2-SAT

Flow network (4)

9. Max-flow & min-cut

- Weak duality of flows and cuts
- Ford-Fulkerson method
- Max-flow min-cut theorem
- Flow Integrality
- Capacity scaling method

10. Bipartite matching

- Reduction from bipartite matching to max flow
- Hall's theorem for perfect matching: max-flow / min-cut thm vs. Dilworth's thm

11. Reductions to max flow

- Circulation with demands: reduction to max flow
- Survey design: reduction to circulation with demands
- Edge-disjoint paths: reduction to max flow
- Network connectivity: duality with edge-disjoint paths

12. Reductions to min cut

- Image segmentation: reduction to min cut
- Project selection: reduction to min cut

- Open pit mining: reduction to project selection

Greedy algorithms (3)

13. Greedy algorithms for simple problems (exchange argument)

- Selecting items to maximize total weights
- Cashier's algorithm for coin changing
- Truck driver's algorithm to select breakpoints

14. Greedy scheduling algorithms

- Interval scheduling ("stays ahead" argument)
- Interval partitioning (structural property of solution)
- Scheduling to minimize lateness (exchange argument)
- Farthest-in-future (FF) policy for offline caching

15. Huffman codes for data compression

- Optimal prefix codes
- Huffman encoding

★ Minimum spanning tree (deferred to MC3202)

- Cut/cycle properties
- Prim/Kruskal's algorithm
- k -clustering
- Matroids

Divide-and-conquer (2)

16. Divide-and-conquer algorithms

- Counting inversions using merge sort
- Closest pair of points
- Karatsuba's algorithm for integer multiplication
- Strassen's algorithm for matrix multiplication

17. Fast Fourier Transform (FFT)

- Convolutions & polynomial multiplications
- Fast discrete Fourier transform
- FFT circuits, quantum circuits for discrete Fourier transform

Dynamic programming (3)

18. Dynamic programming pattern #1

- Binary choice: weighted interval scheduling
- Multi-way choice: segmented least squares
- Additional variable: knapsack (pseudo-polynomial time algorithm)

19. Dynamic programming pattern #2

- DP over intervals: RNA secondary structure, CYK parsing
- Multi-dimensional: edit distance (linear space)

20. DP optimization techniques

- Convex hull trick
- Divide-and-conquer optimization
- Knuth optimization

★ Shortest path (deferred to MC3202)

- Dijkstra's greedy algorithm for single-source shortest-path problem
- Bellman-Ford algorithm, Floyd-Warshall algorithm
- Negative cycles
- Distance vector protocol

Amortized analysis (1)

21. Amortized analysis

- Aggregate analysis vs. accounting method vs. potential method
- Disjoint-set data structure, union-find with path compression

Linear Programming & Convex Optimization (6)

22. Linear programming

- LP in standard form vs. slack form
- Formulating discrete optimization problems as linear programs

23. Simplex algorithm #1

- Pivoting operations, basic/nonbasic variables
- Correctness (optimality, termination)

24. Simplex algorithm #2

- Initial basic feasible solution
- Anticycling rules (Bland's rule, lexicographic rule)
- Worst-case vs. average-case running time

25. LP duality

- Primal LP & dual LP
- Strong LP duality from simplex algorithm
- Separating hyperplane theorem, Farkas' lemma
- Strong LP duality from Farkas' lemma
- Dual simplex (for ILP heuristics)

26. Interior-point method & ellipsoid method for LP (and beyond)

- Logarithm barrier function, central path, Lagrange multipliers

- Path-following methods, step length parameter, convergence analysis
- Reduced KKT system, normal equations
- Optimization to feasibility, geometric divide-and-conquer, shrinking lemma
- Separation oracle, Grötschel-Lovász-Schrijver theorem

27. Heuristics for ILP (Integer Linear Programming) & gradient descent method

- Gomory's cutting plane method
- Branch-and-cut heuristic for (mixed) ILP
- Unconstrained/constrained gradient descent
- Stochastic gradient descent

Subproblem analysis (2)

28. Subproblem analysis #1

- Small vertex covers (fixed-parameter tractability)
- Vertex cover in bipartite graph
- König-Egerváry theorem (from max-flow/min-cut theorem)

29. Subproblem analysis #2

- Independent set on trees (greedy)
- Weighted independent set on trees (DP)
- Circular arc coloring based on interval partitioning (fixed-parameter tractability)

Approximation algorithms (3)

30. Greedy approximation algorithm

- List scheduling for load balancing
- Center selection
- APX-hardness

31. Pricing method & LP rounding

- Pricing method for weighted vertex cover
- LP rounding for weighted vertex cover
- LP rounding for generalized load balancing

32. Fully polynomial-time approximation scheme (FPTAS)

- FPTAS for knapsack (rounding)
- FPTAS for subset sum (trimming)

Randomized algorithms (2)

33. Randomized algorithms #1

- Monte Carlo, RP (Randomized Poly-time)
- Las Vegas, ZPP (Zero-error Probabilistic Poly-time)

- Linearity of expectation (with indicator RVs)
- Contention resolution
- Contraction algorithm for global min cut (Monte Carlo)
- MAX 3-SAT with waiting-time bound (Las Vegas)

34. Randomized algorithms #2

- Universal hashing, cryptographic hashing
- Chernoff bounds (with independent indicator RVs)
- Load balancing
- Derandomization methods

Online algorithms (2)

35. Online algorithms #1

- Competitiveness, strong competitiveness
- LRU for fixed-space online caching

36. Online algorithms #2

- VMIN for variable-space online caching
- Randomized online caching algorithm

Quantum algorithms and quantum-resistant cryptography (4)

37. Quantum computation #1

- Qubits, superposition, measurement (entanglement)
- Quantum register, unitary matrices, quantum operations (quantum gates)
- BQP, quantum vs probabilistic computation

38. Quantum computation #2

- Quantum circuits, boolean circuits as quantum circuits
- Grover's algorithm, Simon's algorithm

39. Quantum algorithms for factoring & discrete logarithm

- QFT (Quantum Fourier Transform)
- Quantum circuits for modular exponentiation
- Shor's order-finding algorithm
- Reducing factoring & discrete logarithm to order finding

40. Lattice-based public-key encryption schemes

- LWE (Learning With Errors) conjecture
- LWE-based public-key encryption scheme
- Module-LWE conjecture
- Kyber (ML-KEM, Module-Lattice-Based Key-Encapsulation Mechanism)
- Lattice problems, worst-case to average-case reduction

5 Lecture Schedule (Tabular)

Lec #	Topics	Categories
1	Poly-time reductions, 3-SAT	Intractability
2	Certifier-based perspective	
3	3D matching, Hamiltonian cycle	
4	Graph coloring, subset sum	
5	NP-completeness	
6	co-NP	
7	PSPACE	
8	SCC/2-SAT, reductions to 2-SAT	2-SAT
9	Max-flow & min-cut, duality	Flow network
10	Bipartite matching, Hall's condition	
11	Reductions to max flow	
12	Reductions to min cut	
13	Exchange argument (matroids)	Greedy strategy
14	Scheduling problems	
15	Optimal prefix codes	
16	Geometric/arithmetic problems	Divide & conquer
17	Fast Fourier Transform	
18	Binary/multiway choice, new variable	Dynamic programming
19	DP over intervals, multi-dimensional	
20	DP optimization (CHT, Knuth, D&C)	
21	Union-find with path compression	Amortized analysis
22	Linear programming	Mathematical programming
23-24	Simplex algorithm for LP	
25	LP duality	
26	Interior-point method, ellipsoid method	
27	ILP (cutting-plane method), gradient descent	
28	Vertex cover (FPT, König-Egerváry thm)	Subproblem analysis
29	Independent set (greedy/DP), circular arc	
30	Greedy (load balancing, center selection)	Approximation algorithms
31	Pricing method, LP rounding (weighted VC)	
32	Fully polynomial-time approximation schemes	Randomized algorithms
33	Monte Carlo (RP) vs. Las Vegas (ZPP)	
34	Linearity of exp, Chernoff, derandomization	Online algorithms
35	Competitiveness, online caching	
36	Randomized online caching	Quantum computation
37	Quantum computation, BQP	
38	Simon's algorithm, Grover's algorithm	
39	Shor's algorithm for discrete log & factoring	
40	Quantum-resistant public-key encryption	
41	<i>Wrap-up for the final exam</i>	

6 Calendar

Mon	Tue	Wed	Thu	Fri
2/23 L1	2/24	2/25 L2	2/26	2/27 L3
3/2 <i>holiday</i>	3/3	3/4 L4	3/5	3/6 L5
3/9 L6	3/10	3/11 L7	3/12	3/13 L8
3/16 L9	3/17	3/18 L10	3/19	3/20 L11
3/23 L12	3/24	3/25 L13	3/26	3/27 L14
3/30 L15	3/31	4/1 L16	4/2	4/3 L17
4/6 L18	4/7	4/8 L19	4/9	4/10 L20
4/13 L21	4/14 <i>Midt.</i>	4/15 <i>Midt.</i>	4/16 <i>Midt.</i>	4/17 <i>Midt.</i>
4/20 L22	4/21	4/22 L23	4/23	4/24 L24
4/27 L25	4/28	4/29 L26	4/30	5/1 L27
5/4 L28	5/5 <i>holiday</i>	5/6 L29	5/7	5/8 SAF
5/11 L30	5/12	5/13 L31	5/14	5/15 L32
5/18 L33	5/19	5/20 L34	5/21	5/22 교류전
5/25 <i>holiday</i>	5/26	5/27 L35	5/28	5/29 L36
6/1 L37	6/2	6/3 <i>holiday</i>	6/4	6/5 L38
6/8 L39	6/9	6/10 L40	6/11 L41	6/12 L42
6/15 <i>Final</i>	6/16 <i>Final</i>	6/17 <i>Final</i>	6/18 <i>Final</i>	6/19 <i>Final</i>